

Geochemical Characteristics, Genesis of Concealed Cu-rich Ore Body in the Jinchuan Deposit, Northwestern China, and Its Prospecting

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Abstract: The Jinchuan deposit is hosted by the olivine-rich ultramafic rock body, which is the third-largest magmatic sulfide Ni–Cu deposit in the world currently being exploited. Seeking new relaying resources in the deep and the border of the deposit becomes more and more important. The ore body, ore and geochemistry characteristics of the concealed Cu-rich ore body are researched. Through spatial analysis and comparison with the neighboring I11 main ore body, the mineralization rule of the concealed Cu-rich ore body is summed up. It is also implied that Cu-rich magma may exist between Ni-rich magma and ore pulp during liquation differentiation in deep-stage chambers, which derives from deep-mantle Hi–MgO basalt magma. It is concluded that the type of ore body has features of both magmatic liquation and late reconstruction action. It has experienced three stages: deep liquation and pulsatory injection of the Cu- and PPGE-rich magma, concentration of tectonic activation, and the later magma hydrothermal superimposition. In addition, the Pb and S isotopes indicate the magma of I6 concealed Cu-rich ore body originates predominantly from mantle; however, it is interfused by minute crust material. Finally, it is inferred that the genesis of the Cu–Ni sulfide deposit is complex and diverse, and the prospect of seeking new deep ore bodies within similar deposits is promising, especially Cu-rich ore bodies.

Key words: Cu–Ni sulfide deposit, Cu-rich magma, Cu-rich ore body, prospecting, Jinchuan

1 Introduction

The Jinchuan deposit, located in Jinchang, Gansu, northwest China, is the world's third-largest exploited magmatic Cu–Ni sulfide deposit, and contains 17 elements, such as Co, Au, Ag, PGE, as well as Cu and Ni (SGT, 1984). The total proven ore and Cu–Ni metallic reserves are respectively 550 million tons of ore, 5.58 million tons of Ni, and 3.54 million tons of Cu. The average contents of Pt, Pd, Au, and Ag are 0.20, 0.12, 0.08, and 3.2 ppm, respectively. It has been prospected and exploited for more than 50 years since its discovery in 1952, and few rich ore bodies have gradually been depleting. The phenomenon of the resource crisis in some segments of the Jinchuan deposit has begun to emerge. I11 main ore body (I11) is the largest rich ore body, and its reserve and content of Ni is 48.4% and 56.8%, respectively, of the whole Jinchuan deposit. Some concealed ore bodies in the deep and the border of the Jinchuan deposit during exploiting and

prospecting have been found, for example, the I6 concealed Cu-rich ore body (I denotes mining area I; 6 is the number of ore bodies), the I11 concealed rich ore body, the I11S8 metasomatic ore body, and I11 massive ore body. Among these, the I6 concealed Cu-rich ore body (I6 Concealed) is a representative study of the Jinchuan Cu–Ni sulfide deposit. I6 Concealed is a new type of Cu-rich ore body, found along the fault and discovered in the course of mining. It contains normal Ni and a rather higher grade of Cu, Au, Ag, and PPGE. By studying its geology and geochemical characteristics, the presence law, genesis, and deep extension of I6 Concealed may be understood and mastered. It is of significant importance for studying the evolutionary history of the Jinchuan deposit, deep prospecting, and for understanding the passage of magma (Tang et al., 2006).

There are many Cu–Ni sulfide deposits in the world similar to the Jinchuan deposit in metallogenic model or ore-controlling conditions, for example, the Huangshandong deposit (China) (Zhu, 1996), the Karatungke deposit (China) (Zou et al., 2000), the Noril'sk

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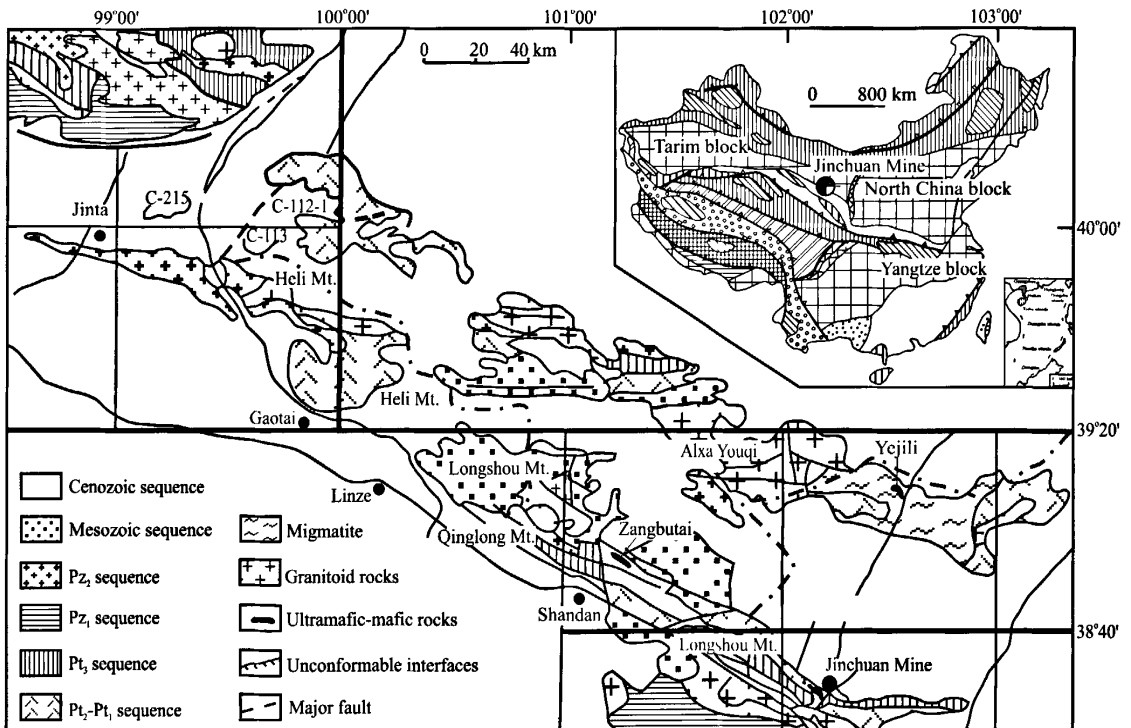


Fig. 1. Simplified geological map of the Jinchuan deposit.

deposit (Russia) (Naldrett, 1992, 1997, 1999; Luo et al., 2000), the Voisey's Bay deposit (Canada) (Naldrett, 1997, 1999; Li et al., 2001), and the Duluth deposit (USA) (Bao and Ripley, 1983; Naldrett, 1997, 1999). The theory and production of studying I6 Concealed might be useful for prospecting in the depth of these deposits.

2 Geological Setting

The Jinchuan ultramafic intrusion is situated on the southwestern margin of Alaxa massif, which intrudes into the Middle Archean metasedimentary rocks of the Baijiazui Formation in the Middle Proterozoic era. The north of the Longshoushan uplift is adjacent to the Chaoshui Depression with the northern fault F1 of Longshoushan (regional fault); its south borders the Qilian Fold Belt with its southern fault (Fig. 1).

The intrusive rock body is approximately 6500 m long, over 20–527 m wide, and more than 1100 m thick. Its strike is N50°W, its trend is SW, and its dip is 40°–70° (SGT, 1984). The east and west ends of the ultramafic intrusion are covered by Quaternary alluvium and the central section is partly exposed, of which the area is approximately 1.34 km². The faults and joints in the mining area are very developed, and are mainly F₁, F₈, F₁₇, F₂₃, F₁₆, F₁₆₋₁, and F₆. The rock body and ore body are comparatively broken. The whole intrusion is divided into four segments by NEE-

trending (NEE, north by east) strikeslip faults. Four segments are respectively referred to as mining area III, I, II, and IV from west to east (Fig. 2). I6 Concealed is in the eastern part of mining area I, adjoining the western end of III main in mining area II.

The Jinchuan rock body is of multiperiod intrusion. The primary intrusive and metallogenetic periods have four stages (Tang and Li, 1995): the primary ultrabasic rock of the first stage is medium-fine grain lherzolite and olivine websterite, the second stage is made up of medium-coarse grain pyroxene peridotite and lherzolite, which make up most of ultrabasic rock in the Jinchuan deposit (up to 67.7 vol%); the third stage is made up of medium-grain dunite, which bears most of the rich ore body of the Jinchuan deposit; and the last stage is made up of the ore pulp of massive ore bodies. The dominant rock type in these stages is lherzolite. Generally, the contact relationship between rocks in the same stage is a gradual transition and the contact relationship in the different stage is mutation. The primary igneous minerals are olivine (Fo86–77, SGT, 1984; de Waal et al., 2004; Li et al., 2004), orthopyroxene (En88–72), clinopyroxene (En45–62, Wo37–45, Fs1–11), plagioclase (An63) and a little amphibole hornblende. The secondary minerals include serpentine, magnetite, tremolite, talc, magnesite, calcite, chlorite, and clinozoisite. The typical hydrothermal minerals, such as calcite and chlorite, occur often in small branching veinlets or at the

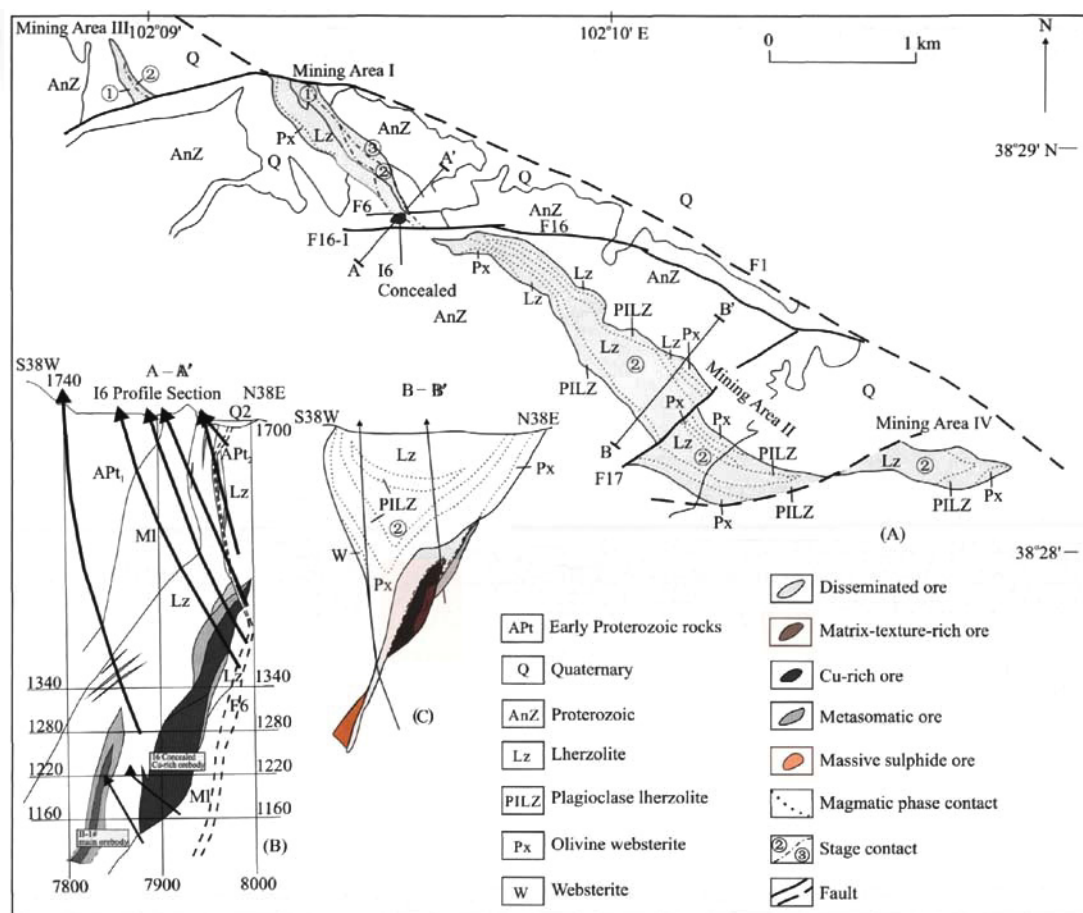


Fig. 2. Geological sketchy plane (A) and sections (B, C) of the Jinchuan deposit (adapted from Tang et al., 2002, with modifications). (B) section of the I6 concealed copper-rich ore body.

termination of the main ore vein.

The ratio of m/f is 3–5.2; the Jinchuan intrusion belongs to the ferrous ultrabasic rock (Yang et al., 1997).

3 Ore Body Characteristics of I6 Concealed

The most important characteristic of I6 Concealed is a rather high Cu content. Its average ratio of Ni/Cu is 1:2.2, and the highest ratio is up to 1:3, whereas the ratio of the Jinchuan general ore body is 2:1, therefore it is the concealed Cu-rich ore body (JNMC, 1997).

The I6 Concealed ore body is located between I5 and I7 (I5 is the number of the exploration line) in the plane and is at a level of 1400–1070 m; the trend is SW (SW, southwest) and the dip is 70°. It hosts in the footwall of fault F6. The maximum thickness of the ore body is over 40 m. It is independent above level 1220 m; but it finally conjoins with I11 below level 1220 m. The phenomenon may imply

the movement direction of the I6 concealed magma is from west to east and from down to up. It also implies the rule of branch integration and bulge shrinkage in I6 Concealed is frequent. Its general spatial configuration is ore polar, so sometimes it is also named I6 Concealed Cu-rich ore polar. Its plane configuration is a little different on different level, for example, the shape of the ore body is lentoid on level 1340 m; the shape of the ore body is the shape of horseshoe on level 1280 m; and the shape of the ore body is stratoid on level 1220 m (Fig. 3).

The I6 Concealed ore body locally contacts with gneiss, amphibolite, marble, granite, and aplite. The contacting borderline between the ore body and wall rock appears obvious. The primary vein rock in the ore body is granite porphyry and lamprophyre. Occasionally, a veinlet of chalcopyrite and pyrite interpenetrates into breccias of lamprophyre. This indicates that I6 Concealed also experienced tectonic reconstruction and ore-forming

hydrothermal activities after the ore body was intruded by lamprophyre.

Fault F6 is situated northwest of fault F16-1, which is one of major faults in the Jinchuan deposit (Fig. 2). Its strike is NEE, its trend NW (NW, northwest), its dip is approximately 70°, and its width is 1–2 m. There appear to be a fault argilla zone, a mylonite zone, and a schisteous lens zone in fault F6. The lens are often brecciate ores, and there have many scratch marks on its surfaces; the side depression angle of scratch marks is 10°–60°. It is judged that F6 is a compresso-shearing slip fault by the sharp angle between the major axis of the lens and the fault plane, and by the side depression angle of scratch marks (Fig. 4A). The primary minerals in F6 are serpentine and chlorite, besides comparatively developing vein carbonate and thin-vein metal sulfides (Lehmann et al., 2007). This indicates that it experienced fault reconstruction and the hydrothermal superimposition after the ore body was injected and mineralized.

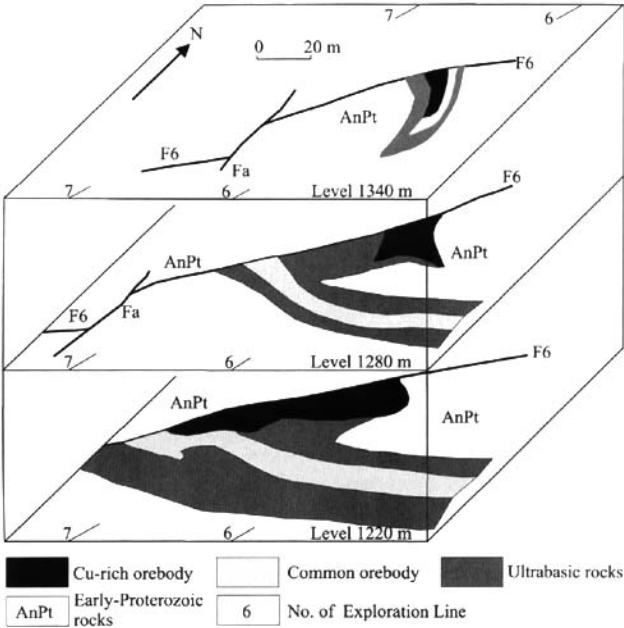


Fig. 3. Tridimensional scenograph of the I6 concealed copper-rich ore body.

4 Ore Characteristics of I6 Concealed

4.1 Lithofacies of ore-bearing and alteration

The primary ore-bearing lithofacies of I6 Concealed are pyroxene peridotite and talc magnesite. Owing to the effect of late hydrothermal superimposition and tectonism and strong alteration, the primary texture and structure of the ore could barely be observed. Moreover, rock-forming minerals and metal sulfides are directionally arranged (Fig. 4B), and the schistosity is highly developed. Olivine and pyroxene are entirely altered into serpentine, chlorite, tremolite, green hydrophlogopite, and iddingsite, among which serpentine, chlorite, and tremolite are common (Rowell and Edgar, 1986; Ripley et al., 2005). After deducting sulfides, the primary lithofacies of the I6 Concealed ore body are found to contain 94.5% olivine and 5.5% pyroxene (prior to alteration). The data differ from the former testing results (Chai and Naldrett, 1992a; Li. et al., 2004), which indicates that the liquation and differentiation of magma is diverse and complex.

4.2 Ore type

The Ni, Cu-rich ore (SNC-A1) of I6 Concealed always distributes in the middle-upper part and the bottom of the ore body. The poor ore (SC-A2) distributes in the middle and southern area of the upper part of

the ore body. The poor ore gradually disappears from a level of 1400 m to 1310 m and the ore turns wholly into the rich ore. Furthermore, it partly turns into the ultrahigh Cu-rich ore (SC-A). Details of the division of ore type are presented in Table 1.

4.3 Ore texture and structure

Generally, the primary ore texture of I6 Concealed is automorphic-allotriomorphic granular texture (medium-coarse granular and inequigranular) and net texture (owing to the exsolution action of the solid solution); the primary ore structure is spongy iron (matrix, Fig. 4C), local spongy iron, and taxitic (belonging to the group of deep liquation and late injection of magma).

4.4 Primary metal minerals and patterns of presence

The primary metal minerals of I6 Concealed contain pyrrhotite, pentlandite, and chalcopyrite. Like the III1 main ore body, pentlandite locally alters into violarite and mackinawite, and pyrrhotite locally alters into pyrite. The

Table 1 Division of ore type in the Jinchuan Cu–Ni sulfide deposit

Ore type	Symbol	Ni content (wt%)	Cu content (wt%)
Ultrabasic Ni–sulfide– poor ore	SN-A2	0.3–1.0	Without limit
Ultrabasic Ni–sulfide– rich ore (Cu<Ni)	SN-A1	1.0–3.0	Without limit
Ultrabasic Ni–sulfide ultrahigh massive ore (Fig.4D)	S-A	?3.0	Without limit
Ultrabasic Cu–sulfide poor ore	SC-A2	<0.30	0.3–1.0
Ultrabasic Cu–sulfide rich ore	SC-A1	<1.0	>1.0
Ultrabasic Ni–Cu–sulfide-rich ore (Cu>Ni) (Fig.4E)	SNC-A1	>1.0	>1.0
Ultrabasic Cu–sulfide ultrahigh ore	SC-A	<3.0	?3.0

Cu, copper; Ni, Nickel; Po, pyrrhotite.

Table 2 Content of primary metal mineral in ores of the I6 concealed copper-rich ore body

Sample code	Primary metal mineral content (wt%)						Total
	Po (py) ¹	Pn (vi) ²	Cp (Cu) ³	Vallerite	Iron oxide	Gangue	
B-14	3.88	2.93	11.06	0.44	7.82	73.87	100
6B-15	5.89	3.48	11.13	1.49	6.54	71.47	100
6B-19	2.95	4.11	13.49	0.41	3.52	75.52	100
6B-16	4.61	3.27	14.71	0.79	10.93	65.69	100
6B-5	13.46	8.46	11.30	0.30	10.42	58.06	100
Average	6.16	4.45	12.34	0.69	7.85	68.92	100

Testing unit: Technique Center of Jinchuan Nickel Mining Corporation, Jinchang, China. Cp, chalcopyrite; Cu, cubanite; Pn, pentlandite, Po, pyrrhotite; Py, pyrite; Vi, violarite. ¹Po bears little py; ²Pn bears little Vi; ³Cp bears little Cu.

Table 3 Content of Au, Ag, and PGE in ores of the I6 concealed copper-rich ore body (I6 Concealed) and its neighboring ore bodies

Code	Sampling position	Sample type	Element content (ppm)						
			Au	Ag	Pt	Pd	Os	Ir	Ru
6B-G1	I6 Concealed, 1220 m, SNC-A1	CS	1.88	23.60	2.10	0.97	0.001	0.0012	0.001
6B-G2	I6 Concealed, 1220 m, SNC-A1	CS	1.38	24.40	1.90	0.34	0.002	0.0015	0.001
6B-G3	I6 Concealed, 1220 m, SNC-A1	CS	2.00	21.90	2.69	0.45	0.001	0.0011	0.001
6B-G4	I6 Concealed, 1185 m, SNC-A1	CS	1.63	28.00	0.46	0.30	0.001	0.0012	0.001
6B-G5	I6 Concealed, 1180 m, SNC-A1	CS	2.55	14.80	1.24	0.28	0.018	0.026	0.008
7B-G1	I7 Concealed, 1190 m, SNC-A1	CS	1.61	18.50	0.18	0.20	0.02	0.028	0.016
7B-G2	I7 Concealed, 1170 m, SNC-A1	CS	1.30	19.30	0.58	0.36	0.007	0.011	0.002
7B-G3	I7 Concealed, 1220 m, SNC-A1	SS	0.34		1.46	0.32	0.002	0.001	0.001
6H-G11	I6 Common, 1280 m, SN-A1	SS	1.45	3.50	0.30	0.32		0.04	
7H-G22	I7 Common, 1235 m, SN-A1	CS	0.17	1.50	0.10	4.00	0.027		0.017
2H-G23	I2 Common, 1180 m, SN-A1	CS	0.16	2.50	0.28	0.59	0.018	<0.01	0.016
7H-G11	I7 Common, 1270 m, SN-A2	SS	0.83	0.75	0.25	0.11		0.01	
	II8-10 III#, 1200 m	Average	0.18	5.05	1.23	0.35	0.021		0.042
	III	Average	0.3	6.1	0.53	0.24	0.025	0.022	0.20
	Massive	Average	0.11	2.66	0.059	0.07	0.031	0.020	0.027
	Noril'sk-1		3.65	0.119	25.103	63.624	0.003	0.008	0.017
	Noril'sk-2		5.37	0.046	105.96	305.84		0.006	0.015
	CI chondrite		0.140	0.199	0.990	0.560	0.486	0.481	0.712

Testing unit: Guilin Mineral Academy, Guilin, China. 6B or 7B samples belong to I6 Concealed; H sample is its neighboring ore body; III is the rich ore of the III main ore body; massive is the average PGE values of ultrahigh Ni-grade ore; Noril'sk-1,2 are two types of Cu-rich ore bodies from Noril'sk, based on Barnes and et Maier (1999); CI chondrite values are also from Barnes and et Maier (1999). Ag, silver; Au, gold; Common, the III main orebody neighboring the I6 concealed Cu-rich ore body; CI chondrite, the first type of carbonaceous chondrite; CS, composite sample; Ir, iridium; Pd, palladium; Pt, platinum; Os, osmium; Ru, ruthenium; SS, single sample.

primary sulfides occur in the way of xenomorphic crystal texture. I6 Concealed contains higher contents of cubanite, vallerite, rare and precious metal minerals, which include hessite, sperrylite, calaverite, and michenerite (Yang et al., 2005); however, chalcopyrite is still the most important Cu-sulfide mineral. The ratio of pyrrhotite, pentlandite, and chalcopyrite of I6 Concealed is 0.50:0.36:1 (Table 2); the ratio of the III main ore body is 3.5:2:1 (the type of deep liquation and injection) or 0.55:0.4:1 (the type of deep liquation and late injection); and the ratio of the massive ore body is 10:3:1 (the type of late injection) (JNMC, 1997). The ratio of I6 Concealed is similar to the ratio of the III main ore body (type of deep liquation and late injection), which implies that the magma source of the two ore bodies are the same.

In the Fe-S (pyrrhotite and pyrite), Ni-Fe-S (pentlandite and violarite), and Cu-Fe-S (chalcopyrite and cubanite) phases figure, most samples of I6 Concealed are located in the range of deep liquation and late injection; only one sample is in the range of late hydrothermal superposition. These results show I6 Concealed is an ore body formed in the later period of the magmatic stage (Fig. 5A). Through the analysis of pyrrhotite, it is indicated that I6 Concealed

experienced magma liquation and late hydrothermal reconstruction (Fig. 4F). It is distinctly different from the ore bodies of hydrothermal superposition (Fig. 5B), which contains more Cu and less Co.

5 General Geochemistry Characteristics of I6 Concealed

5.1 Concentration of primary ore-forming elements

With regard to the eastern common ore body of mining area I (which belongs to III), its grade of SN-A1 is the average of Ni 2.24 wt%, Cu 1.03 wt%, and the grade of SN-A2 is the average of Ni 0.66 wt%, Cu 0.58 wt%. The grade of I6 Concealed is the average of Ni 1.76 wt%, Cu 3.85 wt %, whereas the grade of SN-A1 in its neighboring ore body is the average of Ni 2.54 wt%, Cu 1.09 wt%, and the grade of SN-A2 in its neighboring ore body is the average of Ni 0.62 wt%, Cu 0.65 wt%. These indicate that I6 Concealed generated superimposing action to the neighboring ore body (Chai and Naldrett, 1992b).

According to this (Fig. 6), whether it be in the planes or sections, the Cu content of I6 Concealed is higher than that of other ore bodies and its Ni content changes smoothly.

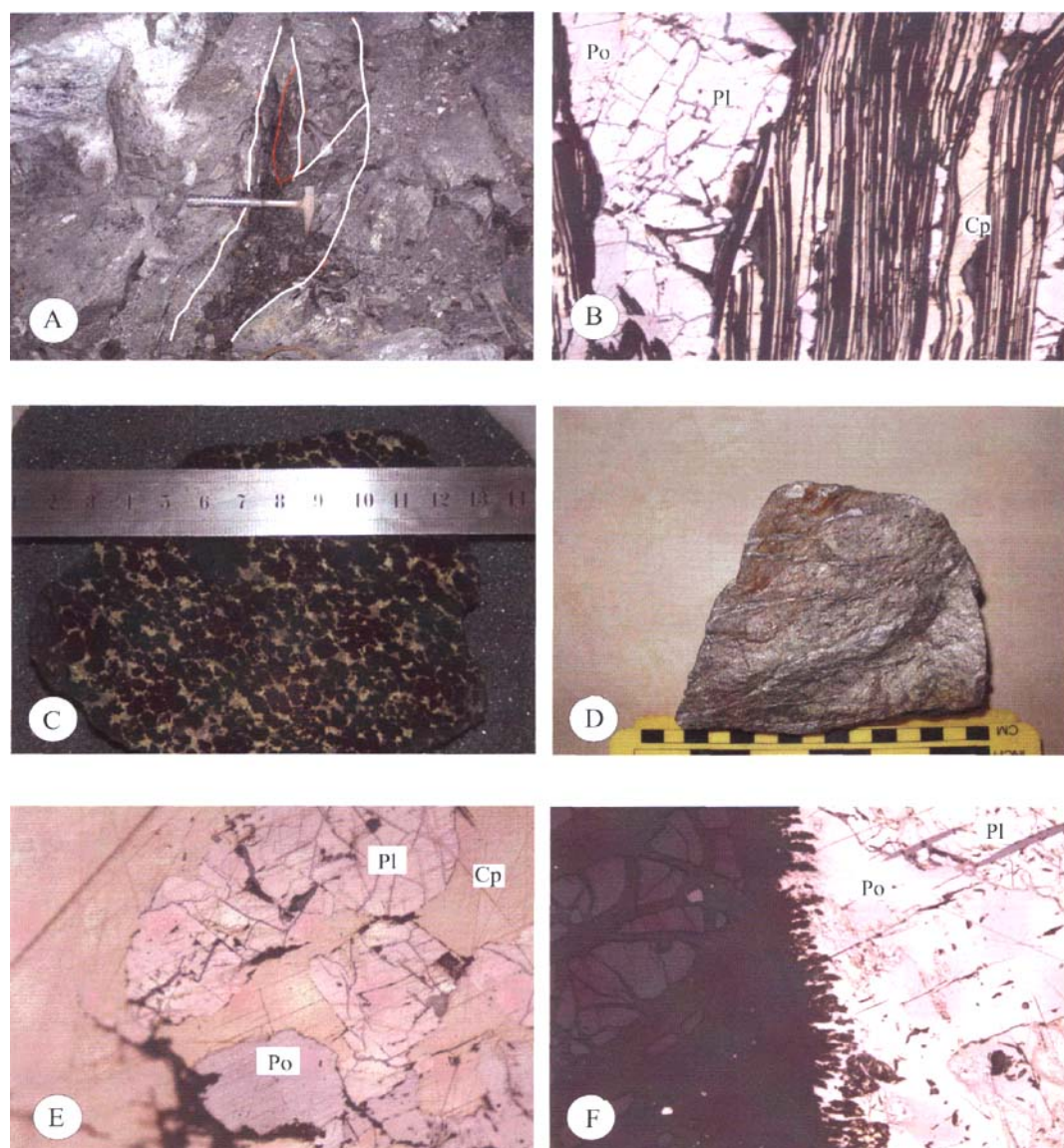


Fig. 4. (A) ore-bearing lens resulting from stress-action; (B) directional arrangement of metal sulfides; (C) matrix structure of the copper-rich ore; (D) massive structure of the nickel-rich ore; (E) paragenetic metal sulfides of the same period; (F) parrhotite melting and alteration by late hydrothermal superimposition. Cp, chalcopyrite; Pl, pentlandite; Po, pyrrhotite.

In planes, the Cu content of I6 Concealed is higher than its Ni content, the Cu content is highest in the central part of I6 Concealed, and in the border contacting with neighboring ore body, the Cu content gradually declines to the general distribution standard of the Jinchuan deposit ($\text{Ni} > \text{Cu}$).

In sections, with I6 Concealed extending deep, its Cu content increases incrementally, the Cu and Ni contents of neighboring SN-A2 do not change much, but the contacting

relationship between neighboring SN-A1 and neighboring SN-A2 is abrupt.

5.2 Primary associated elements

The concentration rule of metallogenetic elements is like that of the similar deposits in the world, and Cu-rich are always Au, Ag, Pt, and Pd rich. The associated elements of I6 Concealed and III are basically similar. They also contain Au, Ag, Co, and PGE, but the content of Cu, Pt, Pd,

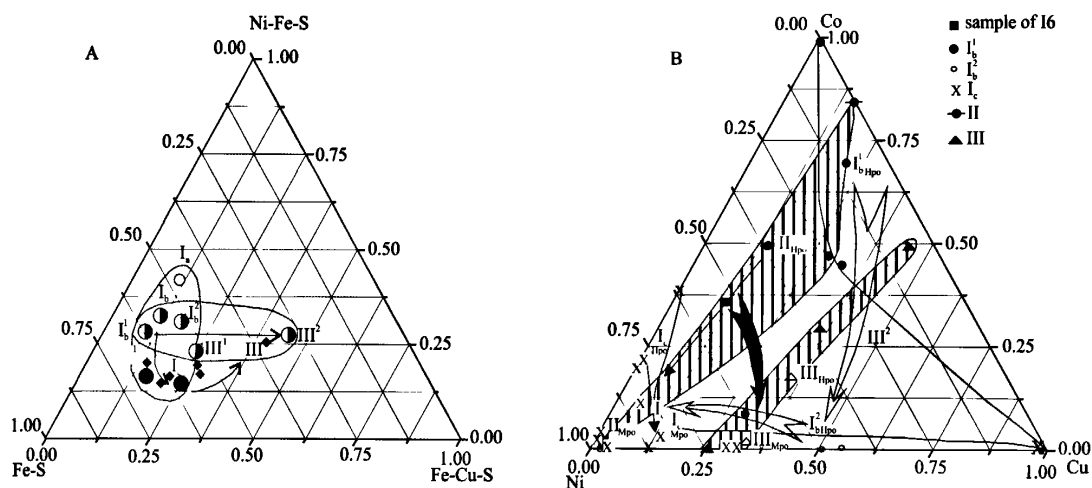


Fig. 5. (A) evolution of metallic minerals of Fe-S, Ni-Fe-S and Cu-Fe-S phases; (B) evolution of the Co, Ni and Cu content in pyrrhotite from various types of ores from the Jinchuan deposit (adapted from Tang and Li, 1995, with modifications).

Hollow arrow shows evolution direction in the magmatic stage, solid arrow is the evolution direction in post-magmatic superimposing stage. I_b^1 , local matrix structure ore (the type of deep liquation injection); I_b^2 , matrix structure ore (the same I_b^1); I_c , massive structure ore (the type of injection during the late stage); II, disseminated structure ore (the type of contacting metasomatic); III, varied matrix structure ore (the type of hydrothermal superimposition); Co, cobalt; Cu, copper; Fe, iron; Hpo, hexagonal pyrrhotite; Mpo, clinopyrrhotite; Ni, nickel; S, sulfide.

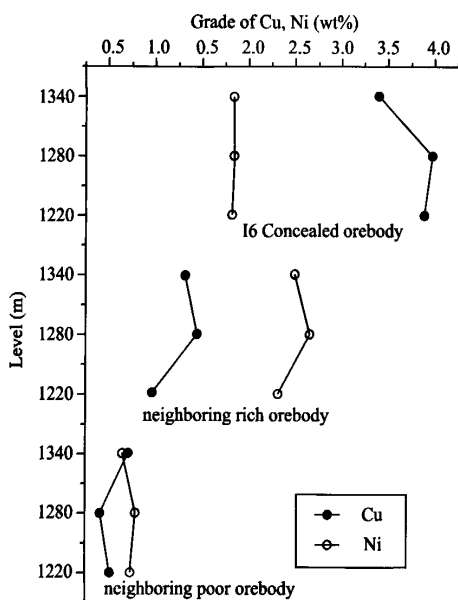


Fig. 6. Spatial distribution of copper and nickel in the I6 concealed Cu-rich ore body and neighboring ore body.

Table 4 Correlation coefficient of six primary metal elements in the I6 concealed Cu-rich ore body

Element	Au	Ag	Cu	Ni	Co	Pt
Au	1.000	0.991	0.903	0.625	0.204	0.778
Ag		1.000	0.948	0.659	0.263	0.841
Cu			1.000	0.698	0.379	0.952
Ni				1.000	0.882	0.518
Co					1.000	0.271
Pt						1.000

Ag, silver; Au, gold; Co, cobalt; Cu, copper; Ni, nickel; Pt, platinum.

Au, Ag, Se, and Te in the I6 Concealed are obviously higher than the ones of the rest ore bodies in the Jinchuan deposit (including III). The precious metal elements in the I6 Concealed ore body, which are exceedingly concentrated, are several times higher than the ones of other ore bodies (Table 3).

According to the equation of Barnes and Lightfoot (2005), the content of samples are converted into the content in 100% sulfides, in which S is 8.65 wt%, Cu is 3.85 wt%, and Ni is 1.76 wt%.

$C_{(100\% \text{ sul})} = C_{\text{wr}} \times 100 / (2.527 \times S + 0.3408 \times \text{Cu} + 0.4715 \times \text{Ni})$ (1) where $C_{(100\% \text{ sul})}$ is the concentration of an element in 100% sulfides, C_{wr} is the concentration of the element in the whole rock; S, Cu, and Ni is the concentration of these elements in the whole rock, in wt percentage.

In the I6 Concealed, Ir is positively correlated with Os and Ru, and negative correlated with Pt, Pd, Au, and Ag (Fig. 7). Its IPGE is close to or slightly lower than those of III SN-A1 (rich ore), higher than those of massive ores, and far higher than those of neighboring hydrothermal superposition ores. Its PPGE, Au, and Ag are close to those of III rich ores, close to or slightly lower than those of hydrothermal superposition ores, and higher than those of massive ores. The positive correlation between Pt and Pd and Au and Ag is quite obvious. In general, the correlation coefficient (r) between Ni and Co is high ($r=0.882$, Table 4), and Cu is strongly correlated with Au, Ag, and Pt ($1 > r > 0.90$) (Wang and Shi, 1999).

IPGE shows that I6 Concealed and III belong to the same type of magmatic liquation. The stage of magmatic liquation differentiation of I6 Concealed is later than that of

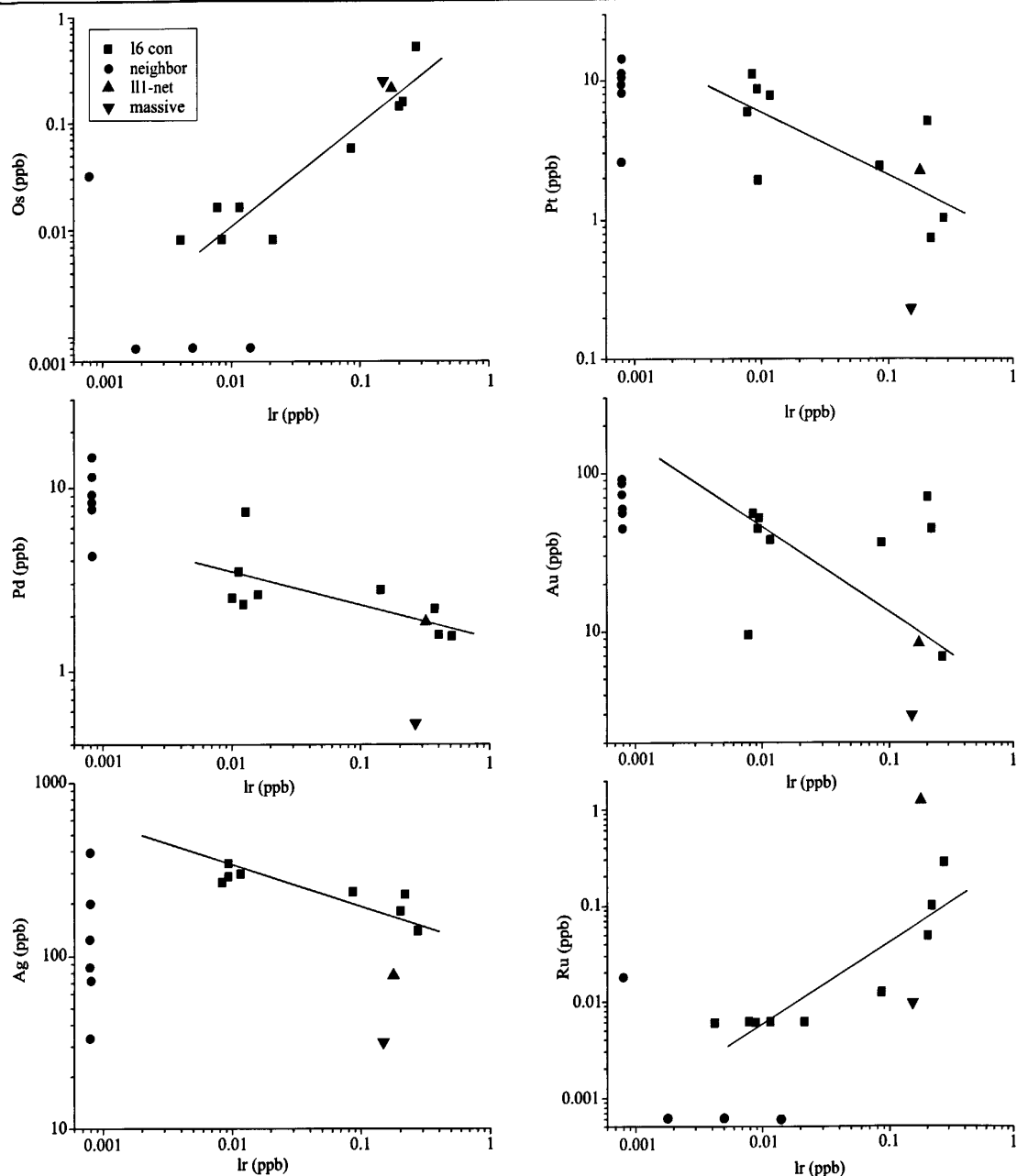


Fig. 7. Relative diagram of PGE–Au–Ag against Ir of the I6 concealed Cu-rich ore body and other ore bodies of the Jinchuan deposit. I6 con, I6 concealed Cu-rich ore body; neighbor, the neighboring ore; III-net, the rich ore of III main ore body; massive, the ultrahigh Ni-grade ore. Ag, silver; Au, gold; Cu, copper; Ni, nickel; PGE, platinum group element.

III, and slightly earlier than that of late-injection massive ore bodies. PPGE indicates that PPGE-rich I6 Concealed had experienced the late intensive hydrothermal process, which caused its PPGE, Au, and Ag to be further concentrated. Yang et al. (2006) considers three types of PGE in the Jinchuan deposit. From the center to the border, the three types of PGE ores can be found in I6 Concealed. It is also implied that I6 Concealed had successively gone

through the processes of magmatic liquation, hydrothermal superposition, and ductile tectonic reconstruction.

The average PGE content of I6 are Au 1.9 ppm, Ag 19.2 ppm, Pt 1.7 ppm, Pd 0.9 ppm, Os 0.005 ppm, Ir 0.01 ppm, and Ru 0.004 ppm. Compared with the neighboring hydrothermal ore, the rich ore of III, the Ni-rich massive ore of the Jinchuan deposit, and the Cu-rich ore of Noril'sk (Barnes and Maier, 1999), the following information may

Table 5 Analysis results of lead isotope in the I6 concealed Cu-rich ore body (I6 Concealed) and neighboring ore bodies

Code	Sample site	Analysis method	Pb isotope ratio			Apparent age (Ma)	Φ value	M value	Th/U
			$^{206}\text{Pb}/^{204}\text{Pb}$	$^{207}\text{Pb}/^{204}\text{Pb}$	$^{208}\text{Pb}/^{204}\text{Pb}$				
6B-5	1220 m, I6 Concealed	WR	16.852±0.011	15.360±0.012	37.071±0.038	812	0.671	8.66	3.99
6B-5		Po	16.924±0.021	15.385±0.029	37.175±0.087	786	0.688	8.70	4.00
6B-11	1280 m, I6 Concealed	WR	17.050±0.002	15.372±0.001	37.116±0.002	675	0.656	8.65	3.89
6B-11		Po	17.031±0.014	15.401±0.013	37.168±0.035	723	0.661	8.71	3.93
6B-13		WR	16.852±0.030	15.414±0.036	37.143±0.100	872	0.679	8.77	4.04
6B-14		WR	16.747±0.089	15.350±0.009	36.790±0.021	881	0.680	8.66	4.00
6B-16		WR	16.813±0.016	15.358±0.016	37.076±0.052	841	0.675	8.67	4.02
6B-22		WR	16.919±0.002	15.349±0.002	37.032±0.005	749	0.664	8.63	3.92
6B-18	Granite	WR	17.420±0.004	15.423±0.003	38.074±0.008	452	0.632	8.70	4.14
7H-20	1270 m, I7 neighboring ore body	WR	17.476±0.012	15.482±0.014	37.673±0.045	480	0.635	8.80	3.92
7H-20		Po	17.509±0.039	15.512±0.038	37.705±0.104	491	0.636	8.86	3.92
7H-20		PYF	17.349±0.032	15.378±0.031	37.343±0.067	452	0.632	8.62	3.82

Testing unit: Beijing Testing and Analyzing Center, Beijing, China. 6B-18 sample belongs to the neighboring granite; 7H-20 samples belong to the neighboring ore body; remaining samples belong to I6 Concealed. Cu, copper; M value, modal value; Ni, nickel; Pb, lead; Po, pyrrhotite; PYF, pyrite film; Th, thorium; WR, whole rock.

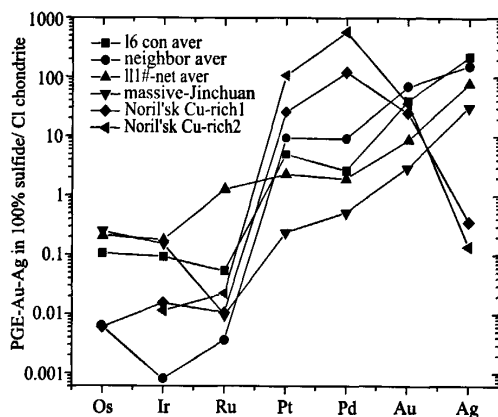


Fig. 8. CI chondrite normalized PGE-Au-Ag patterns of the I6 concealed Cu-rich ore body (I6 Concealed) and other ore bodies of the Jinchuan deposit.

I6 con aver, average PGE grade of I6 Concealed; neighbor aver, average PGE grade of its neighboring ore body; III-net aver, average PGE grade of rich ore of the III main ore body; massive Jinchuan, average PGE grade of the ultrahigh Ni-grade ore; Noril'sk Cu-rich 1, 2, two PGE grade from the Cu-rich ore bodies of Noril'sk, based on Barnes and et Maier (1999). Grade of CI chondrite is also from Barnes and et Maier (1999). Ag, silver; Au, gold; CI chondrite, the first type of carbonaceous chondrite; Cu, copper; Ni, nickel; PGE, platinum group element.

be gained: the PGE, Au, and Ag values of I6 Concealed are between the rich ore of III and the neighboring hydrothermal ore; its IPGE values are two orders of magnitude higher than that of the Cu-rich ore of Noril'sk, and its PPGE values are two orders of magnitude higher than that of the Cu-rich ore of Noril'sk (Fig. 8). This suggests that the I6 Concealed is formed in the later period of the magma stage. To determine which the Cu-rich ore of Noril'sk and the Ni-rich massive ore of Jinchuan are formed in the latest period of the magma stage, from the answer can be found in the different parent magma or different conditions of liquation differentiation in the late stage. Cu is prone to concentration in basic magma, and Ni is prone to concentration in ultrabasic magma. It can be also

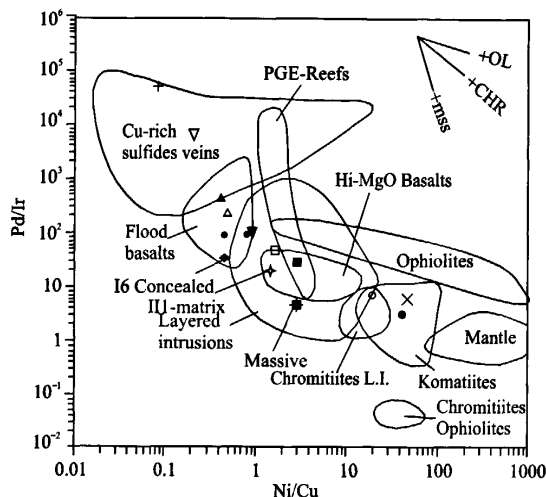


Fig. 9. Ratio plots of Ni/Cu vs Pd/Ir (modified from Barnes and Lightfoot, 2005).

I6 con, the I6 concealed Cu-rich ore body; III net, rich ore sample of the III main ore body; massive, sample of the ultrahigh Ni-grade ore. Perseverance sulfides hosted in komatiite (circles, ○) plot in the komatiite field, Voisey's Bay sulfides (squares, □) plot in the high Mg basalt field, the Noril'sk disseminated and massive sulfides (triangles, ▲) plot in the flood basalt field, the Noril'sk Cu-rich sulfide (inverted triangle open, ▼) and vein (+) plot in the vein field. In general, massive sulfides (solid symbols, ■) have higher Ni/Cu ratios. Cu, copper; Ni, nickel; Ir, iridium; Pd, palladium.

proved that the relationship to the Limahe Intrusion and the Emeishan Basalts (Li et al., 2008). The question will be studied in the future.

The ratio of the $(\text{Pt}+\text{Pd})/(\text{Os}+\text{Ir}+\text{Ru})$ of Cu-Ni sulfide deposit related to basalt magma is between 5.7 and 55.6; its ratio of $\text{Pt}/(\text{Pt}+\text{Pd})$ is approximately 0.2 (Naldrett, 1997). The value of $(\text{Pt}+\text{Pd})/(\text{Os}+\text{Ir}+\text{Ru})$ of I6 Concealed is 47.8, and the value of $\text{Pt}/(\text{Pt}+\text{Pd})$ is 0.76, which are obviously close to the above range. In the ratio plots of Ni/Cu to Pd/Ir (Fig. 9), only the rich ore of III main ore body is in the area of the Hi-MgO basalt magma; neither I6 Concealed nor the massive ore body are in it. This can be explained by Su et al. (2008) who found that part of the ore bodies of the Jinchuan deposit experienced intensive hydrothermal

alteration in the late stage, which could not accurately reflect the characteristics of parent magma in the Jinchuan deposit.

5.3 Pb Isotope research

Doe and Stacey (1974) found that Pb of high μ value is from the lower crust or upper mantle, the μ value of Pb from primitive mantle is between 7.0 and 7.8, the μ value of Pb from the upper crust is above 9.0, and the μ value of Pb from the mixing source is between 7.0 and 9.0 (Albarede and Juteau, 1984). Zhu et al. (2008) confirmed the viewpoint through studying the Bainiuchang Ag-polymetallic Deposit, Yunnan Province, China. Chen et al. (2005) considered the μ value of granulite facies from the lower crust at between 4.0 and 5.0, and its Th/U ratio above 4.5, and the μ value of granite from the upper crust at 15.0, and its Th/U ratio at approximately 3.8. The results of the Pb isotope show the μ value of I6 Concealed is abundant in 8.63–8.77 and its Th/U value

concentrates in the range of 3.89–4.04 (Table 5). This indicates that Pb of I6 Concealed derives from the mixing crust mantle material, in which the mantle derived is predominant and the crust derived is rather little. The characteristics of the Pb samples from I6 Concealed are shown in Fig. 10. The Pb samples from I6 Concealed distributes between the evolution curve of Pb from the island arc and that from mantle, and it much approaches that from mantle. Moreover, the formation age of I6 Concealed is earlier than that of the massive ore body (Fig.10A), the poor ore body, the rich ore body, and the disseminated ore body, which experienced the action of late hydrothermal superimposition (Fig. 10B). Owing to the action of hydrothermal superimposition in the later stage, the samples of its neighboring ore body and granite are located in the range of the late hydrothermal ore body (I, Fig.10B, C). These imply its neighboring ore body is

Table 6 Analysis results of sulfur isotope in metal sulfides of the I6 concealed copper-rich (I6 Concealed) and II1

No.	Sample code	Sulfides	$\delta^{34}\text{S}$ (‰)	Sampling position	Sample characteristics
1	I-TO0402	Chalcopyrite	3.6	Level 1301 m, I6 Concealed, lane 9	Coarse-granular matrix rich ore
2		Pyrrhotite	−0.8		
3	I-TO0423	Chalcopyrite	−1.1	Level 1301 m, I6 Concealed, lane 6	Rich ore
4		Pyrrhotite	−4.0		
5*	II-TO0436	Chalcopyrite	−0.9	Level 1118 m, II1, lane 2	Matrix rich ore
6*	II-O-1	Chalcopyrite	−3.0	Level 1118 m, II1, lane 38	Dunite, rich ore
7*	II-O-7	Chalcopyrite	−3.2	Level 1118 m, II1, lane 22	Rich ore, vein belt

Testing unit: Mineral Resources Institute of the Chinese Academy of Sciences, Beijing, China. *, II1 samples; remaining samples belong to I6 Concealed.

Table 7 S isotope composition of different ore types from the Jichuan Cu–Ni sulfide deposit

No.	Sample code	Sampling position	Ore type	Mineral	$\delta^{34}\text{S}$ (‰)
1	CBZHS-7	II-Line 50, boring hole zk71	Local matrix structure	Po	+2.16
2	CBZHS-4	II-Line 12, boring hole zk22	Matrix structure	Pn	+2.08
3	CBZHS-5-1	II-Line 12, boring hole zk22	Matrix structure	Pn	−0.32
4	CBZHS-5-2	II-Line 12, boring hole zk22	Matrix structure	Py	+2.53
5	CBZHS-6	II-Line 12, boring hole zk22	Matrix structure	Pn	+2.44
6	CNZHS-8	II-Line 50, boring hole zk71	Matrix structure	Py	+1.04
7	CBZHS-10-1	II-Line 8, boring hole zk36	Variant matrix structure	Pn	+2.53
8	CBZHS-10-2	II-Line 8, boring hole zk36	Variant matrix structure	Cp	+2.44
9	CBZHS-11-1	II-Line 14, boring hole zk51	Variant matrix structure	Cp	+2.71
10	CBZHS-11-2	II-Line 14, boring hole zk51	Variant matrix structure	Po	+3.07
11	CBZHS-14	II-Line 36, shaft	Massive	Pn	+1.67
12	CBZHS-15	II-Line 46, boring hole zk98	Massive	Po	+1.85
13	CBZHS-16		Massive	Po	+2.3
14	N063-Po	II-Line 37, Level 1250m	Local matrix structure	Po	+0.05
15	N169-Po	II-Line 38, Level 1250m	Matrix structure	Po	+0.33
16	N060-Po	II-Line 35, Level 1250m	Massive	Po	+0.19
18	N060-Pn	II-Line 43, Level 1250m	Massive	Pn	−1.06
19*			Massive	Cp	+0.21
20*			Massive	Po	−0.3
21*			Massive	Po	−0.75
22*			Massive	Po	+0.68
23*			Massive	Cp	+0.29
24*			Massive	Cp	−0.73
25*			Dunite rich ore	Cp	+1.53
26*			Dunite rich ore	Po	−0.06
27*			Pyroxene peridotite rich-ore	Po	+0.48
28*			Pyroxene peridotite rich-ore	Po	+0.39
29*			Pyroxene peridotite rich-ore	Po	+0.2
30*			Pt-bearing rich-ore	Po	−0.21
31*			Pt-bearing rich-ore	Pn	−0.36
32*			Pt-bearing rich-ore	Pn	+0.45

CBZHS samples are from the SGT (1984); N samples are from Yang et al. (1997); * samples are from Liang et al. (1997). Cu, copper; Cp, chalcopyrite; Ni, nickel; Pn, pentlandite; Po, pyrrhotite.

hydrothermally enriched after I6 Concealed injecting. The $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ correlation of samples from I6 Concealed is good, which is similar with the correlation of samples from II1 (Fig. 10D). The cross-point of the correlation trend line between I6 Concealed and the neighboring ore body possibly denotes the age of the hydrothermal action in the later stage.

The results of apparent age show its primary apparent age of samples distribute between 675 and 881 Ma. Its average apparent age is 792 ± 27 Ma (mean squared weighted deviates [MSWD] is 21.2). It is earlier than those of neighboring granite (452 Ma) and the hydrothermal ore body (452–491 Ma). Although its Th/U value is concentrated, its MSWD is slightly large. This implies that the ore-forming system of I6 Concealed is not completely closed, and it is contaminated by very little radioactive Pb. So the dating data need to be corrected. Meanwhile, it may

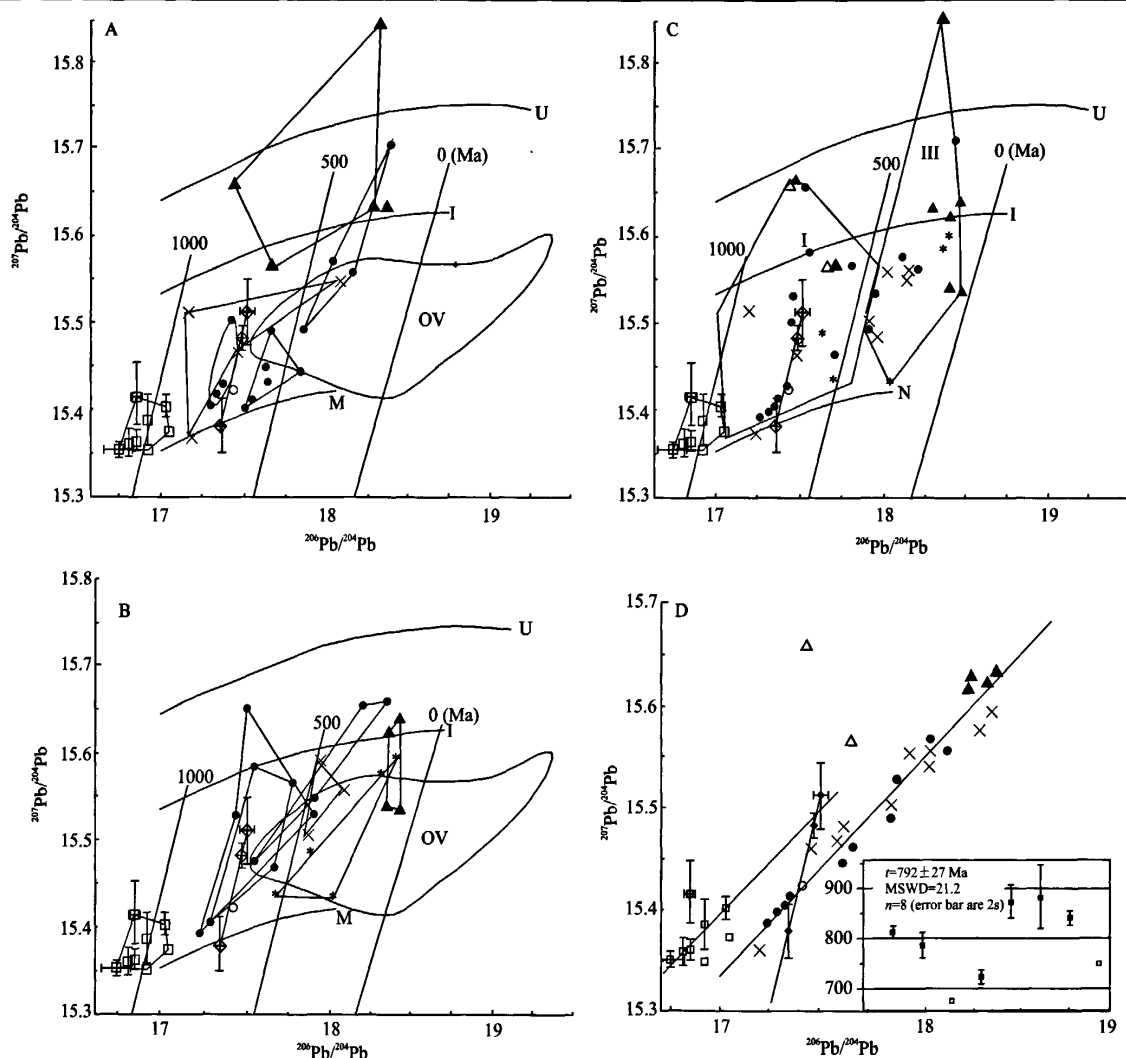


Fig. 10. (A) $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ of I6 concealed Cu-rich ore body (I6 Concealed) and massive ore body; (B) $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ of I6 Concealed ore body and rich ore, poor ore, and disseminated structural ores of the hydrothermal reconstruction ore body; (C) $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ ore age; (D) Correlation figure of $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ ores.

I, evolution curve of Pb from the island arc; IM, evolution curve of Pb from mantle; OV, evolution curve of Pb from ocean volcanic rock; U, evolution curve of Pb from upper crust; (□) I6 Concealed samples; (◇) neighboring ore body samples; (○) neighboring granite samples. Remaining symbols belong to other ore types (from the Jinchuan Nonferrous Metal Corporation). Cu, copper; Pb, lead.

provide evidence that I6 Concealed and the neighboring ore body have suffered from the Jinning Movement and the Caledonian Movement after their injection.

In addition, through the Holmes–Houtermans equation, the formation age is believed to be in 800–1000 Ma. The data is the result of error correction. Compared with the method of U–Th–Pb, when the Pb–Pb dating result is between 0.5 and 1.0 Ga, its error is ~ 250 Ma; when the Pb–Pb dating result is between 1.0 and 2.0 Ga, its error is ± 100 Ma; when the Pb–Pb dating result is between 2.0 and 3.0 Ga, its error is ± 250 Ma (Doe and Stacey, 1974). The result is shown in Fig. 10A–C and is close to the previous dating data of the Jinchuan deposit. These findings can also

indicate that the formation age of I6 Concealed is likely earlier than those of the massive ore body and the hydrothermal ore body (Keays et al., 2004).

5.4 S Isotope research

S is the necessary mineralizer in the course of forming magmatic deposit. Its isotope is a useful indicator by which mineralization is studied and the source of S is judged. According to the data of thermodynamics and isotope fractionation, Ohmoto (1972) indicated that the composition of S isotope in minerals is controlled by t , pH, f_{O_2} , I (ionic strength of alkaline metal in solution), and $\delta^{34}\text{S}_{\text{ES}}$ (average isotope composition of total S in solution):

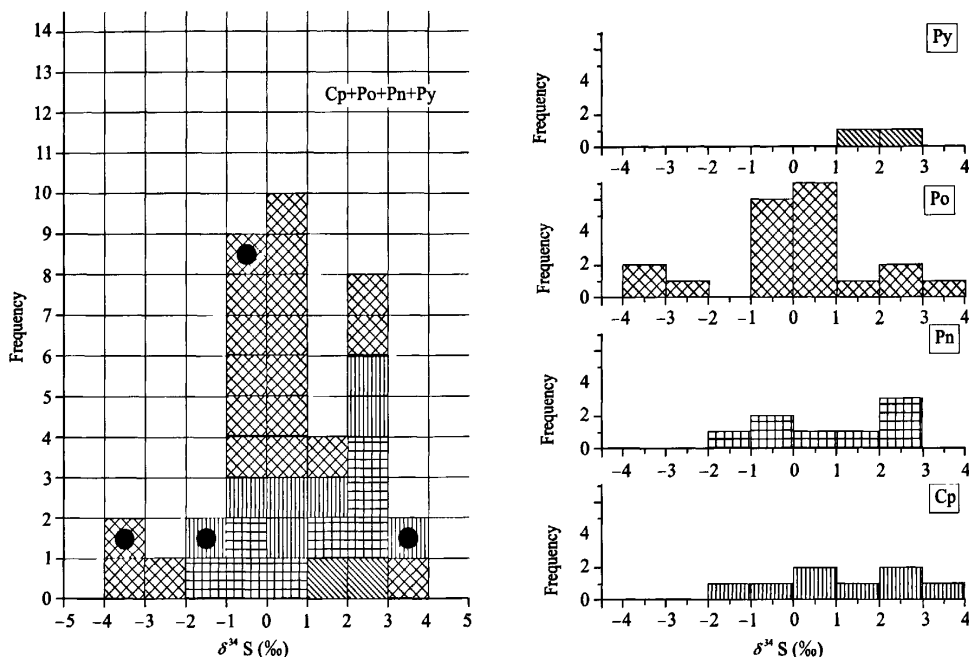


Fig. 11. Stacked histogram of $\delta^{34}\text{S}\text{‰}$ values from the various types of minerals in the Jinchuan deposit. (A) mixture of chalcopyrite (Cp), pyrrhotite (Po), pentlandite (Pn), and pyrite (Py); (B) metallic minerals (Cp, Po, Pn, and Py). (Red circle) samples of the I6 concealed Cu-rich ore body. Remaining samples are from SGT (1984), Yang et al. (1997), and Liang et al. (1997). Cu, copper.

$$\delta^{34}\text{S}_i = f(\delta^{34}\text{S}_{\text{ES}}, t, f_{\text{O}_2}, \text{pH}, I) \quad (2)$$

With decreasing temperatures, S of the oxidized style gradually increases, S of the low valence stepwise reduces, and the $\delta^{34}\text{S}$ value incrementally increases (Coleman, 1977; Smith, 2000).

Because S data are scant in this research (Table 6), the previous S isotope testing results of I1 and the massive ore body in the Jinchuan deposit are systematically collected (SGT, 1984; Yang, 1997; Liang et al., 1997) (Table 7). The total data of $\delta^{34}\text{S}$ are classified, and the stacked histogram of the $\delta^{34}\text{S}$ value is compiled (Fig. 11). The $\delta^{34}\text{S}$ values in sulfides of the Jinchuan deposit are generally between -4.0‰ and $+4.0\text{‰}$, among which there are two peak values. One is -1.0‰ and $+1.0\text{‰}$ (approaching the $\delta^{34}\text{S}$ value of mantle), and the other is $+2.0\text{‰}$ and $+3.0\text{‰}$. Its distribution is concentrative (Ripley, 2005). The statistical results show that the primary sulfides of the Jinchuan deposit are mainly pyrrhotite, pentlandite, and chalcopyrite, which are formed at higher temperatures; the fractionation effect of the S isotope is rather little under such high temperatures. Among them, the crystallization temperature of primary pyrrhotite is the highest, up to 1190°C (Tang et al., 1995). The reason for the second peak value is perhaps due to the late hydrothermal action and S of high valence. S of high valence gradually increases with decreasing temperatures. The result also indicates that the major S source of metallic sulfides are from mantle (primary S), yet is likely to be

joined by external S (derived from crust) in the late hydrothermal stage.

However, the $\delta^{34}\text{S}$ value of I6 Concealed distributes widely, among which the $\delta^{34}\text{S}$ values of two samples are -4.0‰ (pyrrhotite) and $+3.6\text{‰}$ (chalcopyrite). These findings support that magma of I6 Concealed belongs to the type of late liquation differentiation, and its volatile component and S of the low valence (H_2S) are high. Under high temperatures in the early stage, part S of pyrrhotite is mixed with S of low valence, which leads to a decrease of its $\delta^{34}\text{S}$ value. Under relatively low temperatures of the late stage, violent hydrothermal superposition likely causes increasing of part of the $\delta^{34}\text{S}$ value in chalcopyrite. The $\delta^{34}\text{S}$ values of its rest sample are consistent with the average $\delta^{34}\text{S}$ value of the Jinchuan deposit. So the major S source of I6 Concealed is from primary mantle S. Meanwhile, it is possible that minute S derived from the crust is joined (de Waal et al., 2004; Lehmann et al., 2007). The data of rare earth could also support this conclusion (Yang et al., 1991).

6 Analysis of Genesis of I6 Concealed

Although there have been many research results and theses on the metallogenic model of the Jinchuan Cu-Ni sulfide deposit, there is diversity among viewpoints (Naldrett, 1997, 1999; Zhou et al., 2002; Li et al., 2004; de Waal et al., 2004; Yang et al., 2005; Li et al., 2005; Song et

al., 2006, in press; Lehmann et al., 2007). Tang's theory is one of the views that is very much accepted (Tang and Li, 1995; Tang and Bai, 1999; Tang et al., 2002). He considers that Hi-MgO basalt magma from deep mantle intrudes upward into the deep-crust chamber along the deep fault, which then forms the stratification pattern of barren magma, ore-bearing magma, ore-rich magma, and ore pulp from top to bottom. Finally, they are injected orderly into the present locations to form rocks or deposits under pulsatile tectonic stress.

However, the process of magma deep liquation and upward differentiation is successive. With the geophysical and geochemical condition changing, and with the sulfur solubility altering from unsaturated through saturated and to supersaturated, various magma combinations in different stage chambers generate (Barnes and et Maier, 1999; Barnes and Tang, 1999). According to Kullerud's (1969) experiment of the Ni-Fe-S and Cu-Fe-S system-phase diagrams (Zhao and Zhang, 1987), Mss (monosulphide solid solution) can contain the whole Ni and 2% Cu above 600°C (Fig. 12A). When the temperature drops below 600°C, the solid solution decomposes to generate the combination of chalcopyrite and/or pyrite, or the combination of chalcopyrite and/or pentlandite (Fig. 12B, C). When sulfur fugacity and Cu content in the system increases, the solid solution would dominantly decompose to generate the combination of chalcopyrite and pyrite. With temperatures declining, it is implied that the ore-bearing solution would appear in the transitional process from Ni-rich to Cu-rich and from protosulfide to high sulfide (the crystal order of primary metal sulfide: Mss, Iss [insoluble solid solution], pentlandite, chalcopyrite). Based on I6 Concealed and other new ore bodies found in the deep and the border of the Jinchuan deposit, we may conclude there is Cu-rich magma between Ni-rich magma and ore pulp, and its stage of emplacement is also between that of Ni-rich magma and ore pulp.

In addition, the studies of inclusion and isotope can support findings. I6 Concealed belongs to the ore body of deep liquation and late injection (Yang et al., 2005). Its up-intruding process occurs under the condition of incomplete consolidation of prophase ore-rich magma, and slightly earlier than that of late ore pulp. The geophysical and geochemical conditions during the period are very beneficial for the concentration of Cu, Se, Te, Pt, Pd, Au, and Ag, which transfer from the lower layer of the ore pulp to its upper layer in the stage chambers and are helpful in the formation of Cu-rich magma. Therefore, it leads to the loss of Cu, Au, Ag, and PPGE in the late-injecting massive ore body.

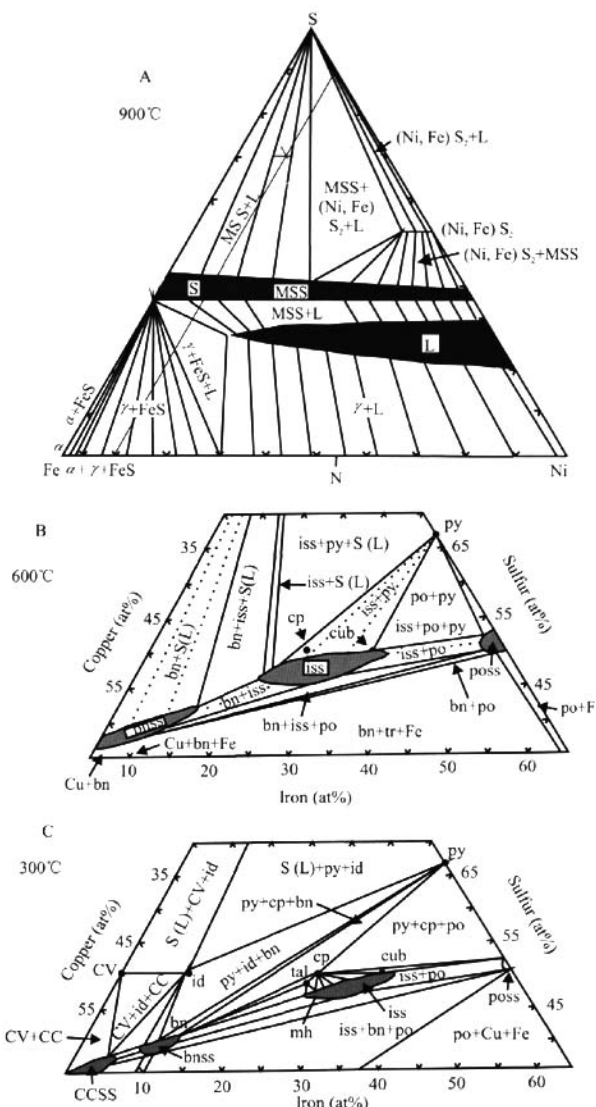


Fig. 12. (A) Phase diagrams of the S-Fe-Ni at 900°C; (B) Phase diagrams of the S-Fe-Cu at 600°C; (C) Phase diagrams of the S-Fe-Cu at 300°C. (adapt after Kullerud et al., 1969).

bn, bornite; cc, chalcosine; cp, chalcopyrite; Cu, copper; cub, cubanite; cv, covellite; Fe, iron; Iss, insoluble solid solution; L, liquid; Mss, monosulphide solid solution; Ni, nickel; po, pentlandite; py, pyrite; S, sulfur; S(L), solution.

The late residual solution of ultrabasic magma contain higher S and volatile components. When magma intrudes, PGE would change from siderophile to strongly sulphophile and chalcophile (the chemical activity of Os, Ru, Rh, Ir, Pt and Pd is orderly increased). Consequently, PPGE (Pt, Pd) in the I6 Concealed ore body is highly concentrated. The IPGE (Os, Ir, Ru, Rh) precipitate earlier, coexisting with pyrrhotite and pentlandite; PPGE, Au, and

Ag precipitate later, coexisting with chalcopyrite, pentlandite, cubanite, and pyrite (Fig. 4E).

Since the Jinchuan deposit is located in the southwestern margin of northern China, it has experienced multistage tectonic movements and frequent conversion between tension stress and compressive stress (Tang and Bai, 1999; Tang et al., 2002; Tu, 2000; Zhai, 2003). The ore-bearing solution transfers from a high-pressure compressive zone to a low-pressure dilatancy area under the effect of the pulsate tectonic pressure gradient. Under the conditions of shear-pressure stress, the ore-bearing solution is activated. Under shear-tension stress, the magma solution then precipitates to mineralize (Zhai and Lü, 2002; Lü, 2003). The pulsatile intrusion of the magma solution is generally along the bottom of the rock body or fault belt formed in the earlier stage. The reason is that these positions are always weak and soft zones, which is advantageous for intruding magma.

The I6 concealed Cu-rich ore body controlled by structure is also one of the outstanding characteristics. Its place of hosting ore is located in the tension fracture zone. The ore-polar configuration of I6 Concealed is obviously controlled by two shear-tension faults with strikes of NEE and NNE. In particular, the NEE F6 is considered to be the passage by which ore-bearing magma injects the present site from depth. Before the neighboring ore bodies are incompletely consolidated along developed faults, the Cu- and PPGE-rich magma intrude upward to form the concealed Cu-rich ore body.

After ore body injection, F6 continues to be active, and its hanging wall transfers upward and eastward. The result causes its primary ore body to rotate clockwise and form the polar ore body. In the course of rotation, the primary ore body would be twisted, squeezed, and crushed. Its part ore-forming elements are again activated and transferred into the squeezed-crushed belt, which causes the existence of a concentrated belt (several meters width) of ultrahigh-grade Cu, Au, Ag, and PGE in the central part of I6 Concealed.

Owing to the intensive alteration of ores in I6 Concealed, its mineral combination and mineral structure is greatly changed. Its primary gangue minerals are mainly altered into serpentine, chlorite, and tremolite. Talc magnesite ore near marble develops, in which there appears lots of chalcopyrite and cubanite. This indicates that after the I6 Concealed polar ore body forms under tectonic activity, it still experiences the superimposing and concentrating effect of hydrothermal fluid, which causes the further enrichment of Cu, Pt, Pd, Au, Ag, and Te. Likewise, it leads to the further enrichment of the neighboring ore body (Tu, 1975; SGT, 1984; Rowell and Edgar, 1986; Ripley et al., 2005).

7 Conclusions

The grade of Ni in the Jinchuan deposit is overemphasized and its associated Cu and PGE have not been paid enough attention. If the Cu grade of deposit is classified like the standard of the Ni grade, I6 Concealed may be an ultrahigh Cu-grade ore body, and the former ultrahigh-grade ore body should be referred to as ultrahigh Ni-grade ore body (Table 1). The two types of ultrahigh-grade ore bodies inject in the last two stages.

A Cu-rich magma between Ni-rich magma and ore pulp during liquation differentiation in deep-stage chambers may exist. The emplacing stage of Cu-rich magma should be between that of Ni-rich magma and that of ore pulp.

The data of Pb and S isotope indicate that the magma of I6 Concealed originates dominantly from mantle; however, it is interfused by minute crust material during emplacement.

I6 Concealed has the features of both magmatic liquation and late reconstruction action, which has experienced three stages: deep liquation and pulsing injection of the Cu- and PPGE-rich magma, concentration of tectonic activation, and the later magma hydrothermal superimposition.

Because the Jinchuan deposit experienced multistage tectonic movements, its ore bodies may be dramatically damaged and reconstructed, and some new types of ore bodies near the main ore bodies are generated, such as the concealed Cu-rich polar ore body, the contact metasomatic ore bodies, the en echelon arrangement little ore bodies, and the ultrahigh-grade disseminated/massive ore bodies. These imply that tectonic movement would further cause diversity and complexity of deposits. Therefore, the prospect of seeking new ore bodies in the deep and the border of the Jinchuan deposit is very promising, in particular, Cu-rich ore bodies.

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